A Computer-Vision-Based Bikeability Score

Bünyamin Aydemir, Gonzalo Guerrero Salazar, and Hendrick Fischer

***Abstract*—**

***Index Terms*—Bikeability, computer vision, zero-shot classification, vision-language models, CLIP, surface classification, object detection, GPX.**

I. Introduction

II. Research Approach and Methodology

III. Objectives and Constraints

Manuscript created July 2026. The authors are students at the Baden-Württemberg Cooperative State University (DHBW) within the Computer Vision course.

IV. Ground Detection: Zero-Shot Classification of the Road Surface

Ground detection determines the ridden surface class for every video frame and thereby provides the first of the three sub-components of the bikeability score. Since each frame corresponds to exactly *one* surface class, the task is formulated as *whole-image classification* (following the principle "whole image = one class") rather than semantic segmentation. The input images originate from a camera mounted at the front of the bicycle handlebar and are sampled at one frame per second.

A. Task Type and Model Class

Because neither annotated target classes nor a project-specific training set are available in field operation, the surface is determined *training-free* (zero-shot). This leads to the choice of *vision-language models* (VLM) of the CLIP family [1], which select the matching class from an image and a set of text prompts without any fine-tuning. Classical CNNs trained on ImageNet do not know application-specific classes such as "gravel path" and are therefore ruled out.

A distinction that is frequently confused in the current literature is central here: *linear probing* of self-supervised features (e.g., DINOv2/DINOv3 [2]) is *not* a zero-shot method. A linear classifier on frozen features requires trained labels of the target classes and is therefore *supervised*. Such a method is deliberately not counted as zero-shot here; at most it could serve as an explicitly declared *non-zero-shot reference point*.

B. Surface Classes and Prompt Ensembling

Four cycling-relevant classes are distinguished: **Cycleway** (paved bike path), **Road** (paved/asphalt road), **Gravel** (gravel/crushed stone), and **Unpaved** (dirt/field track). Since *Cycleway* and *Road* both consist of smooth asphalt on the surface, the text prompts deliberately emphasize the *distinguishing context* (narrow, structurally separated, bicycle pictogram vs. wide, cars, center line, multiple lanes) rather than only the texture. For each class, several English descriptions are used, and their text embeddings are averaged and normalized (*prompt ensembling* as in the CLIP paper [1]), which makes the class embeddings more discriminative. An image is classified via the cosine similarity to these class embeddings; a softmax yields class probabilities.

C. Compared Models

To ensure a fair comparison, all models are accessed through the unified `open_clip` API [3]. A range from *very light* to *medium-sized* is deliberately considered – all local/edge capable, no closed "killer model". Table I summarizes the considered architectures.

Table I
Models compared in the zero-shot benchmark.

Model	**Characteristic**
CLIP ViT-B/32, B/16, L/14 [1]	Classical reference
OpenCLIP (LAION-2B, DataComp-XL) [3], [4]	Dataset influence
SigLIP ViT-B/16 [5]	Efficient zero-shot standard
MobileCLIP S0, S2 [6]	Latency-optimized
EVA02-B/16 [7]	Open zero-shot default

D. Metrics

The benchmark evaluates the models along two axes: *accuracy* and *latency*.

Figure 1. Accuracy (macro-F1) vs. latency (ms/frame, batch size 1). Point size $\propto$ number of parameters; the dashed line marks the Pareto front. *Figure to follow.*

Figure 2. Row-normalized confusion matrices per model. They show which surfaces are confused (typically Gravel $\leftrightarrow$ Unpaved as well as Road $\leftrightarrow$ Cycleway). *Figure to follow.*

a) Accuracy: Since the classes are imbalanced, in addition to accuracy we mainly report the *balanced accuracy* (mean of the per-class recalls) and the *macro-F1*. In addition, a *confusion matrix*, a *bootstrap confidence interval* (95 %, 2000 resamples) on the accuracy, and the *McNemar test* for pairwise model comparison are computed per model. The McNemar test checks on the discordant predictions whether an accuracy difference is *statistically significant* ($p < 0.05$) and not merely noise on a small test set.

b) Latency: On fixed hardware, the pure inference latency is measured at *batch size 1* (real use case: video at the handlebar): several warm-up runs, followed by N measured single inferences including GPU synchronization. The mean, standard deviation, the 95th percentile, and the throughput in frames per second (FPS) are reported.

E. Result Presentation

The central overall representation is a scatter plot of *accuracy vs. latency* (Fig. 1). Each point is a model (point size $\propto$ number of parameters); the drawn *Pareto front* connects the models that achieve the best accuracy for a given latency – i.e., the best trade-off for use at the bicycle handlebar. In practice, **MobileCLIP** variants often lie on the Pareto front, as they offer a very favorable latency-accuracy ratio without requiring an oversized model.

For reproducibility: latency values are strongly hardware-dependent, therefore the concrete computing device (CPU/GPU model, RAM) as well as the used `DEVICE` are documented. All models use the same full image, the same prompt ensembling, and the same `open_clip` pipeline; observed differences therefore stem from architecture and training data, not from the evaluation.

V. Object Detection

VI. Environment Detection

VII. Computation of the Bikeability Score

The bikeability score combines the three sub-components – surface (ground detection), environment (environment detection), and objects (object detection) – into a single value in the interval $[0, 100]$. Each model output is first transformed into a *sub-score* in $[0, 1]$; the three sub-scores are then combined in a weighted manner.

A. Overall Formula

For each evaluated frame, the score A results from the weighted mean of the three sub-scores:

$$A = 100 \cdot \Big(w_g \, S_{\text{ground}} + w_e \, S_{\text{env}} + w_o \, S_{\text{object}} \Big), \tag{1}$$

with the weights

$$w_g = 0.20, \quad w_e = 0.45, \quad w_o = 0.35, \quad w_g + w_e + w_o = 1. \tag{2}$$

Since all sub-scores lie in $[0, 1]$ and the weights sum to 1, the score is *guaranteed* to be bounded to $[0, 100]$.

B. Ground and Environment Sub-Score

The surface and environment sub-scores result from a direct lookup of the respective class quality. If, instead of a single class, a share distribution $\{(k, p_k)\}$ is present, the weighted mean of the class qualities is formed:

$$S_{\text{ground}} = \sum_k q_k^g \, p_k, \qquad S_{\text{env}} = \sum_k q_k^e \, p_k, \tag{3}$$

where p_k is the share of class k ($\sum_k p_k = 1$) and $q_k^g, q_k^e \in [0, 1]$ are the stored quality values. Table II lists the (heuristic) quality values used.

C. Object Sub-Score

The object sub-score evaluates the detected object counts n_k (e.g., cars, buses, cyclists). Via a *saturation function*, a weighted object density is mapped onto $(0, 1]$:

$$S_{\text{object}} = \exp\Big(- \max\left(\rho, 0\right) \Big), \qquad \rho = \frac{1}{N} \sum_k \lambda_k \, n_k, \tag{4}$$

with the number of evaluated frames N and the penalty factors λ_k from Table II. Positive penalty factors (e.g., cars, buses) lower the sub-score, while bicycle-friendly objects such as other cyclists receive a *negative* factor and raise the score slightly. The exponential form is robust against outliers: a single overcrowded frame does not dominate the overall score.

Table II
Class quality values for the ground and environment sub-score as well as object penalty factors.

Component	**Class**	**Value**
Ground q_k^g	Cycleway	1.0
	Road	0.7
	Gravel	0.3
	Unpaved	0.1
Environment q_k^e	Vegetation	1.0
	Water	1.0
	City	0.4
Object penalty λ_k	Car	$+1.0$
	Bus	$+2.0$
	Bicycle	-0.2

Figure 3. Bikeability score colored along the GPX route. The frame scores are merged with the GPX track points via the absolute timestamp. *Figure to follow.*

D. Route Evaluation and Aggregation

To evaluate a complete ride, the video is sampled at fixed intervals (every 5 s, i.e., every $\lfloor \text{FPS} \cdot 5 \rfloor$-th frame) and a score is computed for each frame according to Eq. (1). Each row is stored in a CSV with a timestamp (seconds from video start, `HH:MM:SS`, and optionally an absolute ISO timestamp). Via the absolute timestamp, the frame scores can then be *merged* with the GPX track points of the ride and located geographically. The *average* bikeability score of the route results as the mean over all M evaluated frames:

$$\bar{A} = \frac{1}{M} \sum_{i=1}^{M} A_i. \tag{5}$$

Figure 3 shows the final result: the bikeability score colored along the GPX route, which visualizes the section-wise cycling-friendliness of the ridden route.

VIII. Conclusion

References

[1] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Machine Learning (ICML)*, 2021, pp. 8748–8763.

[2] M. Oquab, T. Darcet, T. Moutakanni *et al.*, "DINOv2: Learning robust visual features without supervision," *Trans. Machine Learning Research (TMLR)*, 2024.

[3] G. Ilharco, M. Wortsman, R. Wightman, C. Gordon, N. Carlini, R. Taori, A. Dave, V. Shankar, H. Namkoong, J. Miller, H. Hajishirzi, A. Farhadi, and L. Schmidt, "OpenCLIP," Zenodo, 2021, https://github.com/mlfoundations/open_clip.

[4] S. Y. Gadre, G. Ilharco, A. Fang *et al.*, "DataComp: In search of the next generation of multimodal datasets," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[5] X. Zhai, B. Mustafa, A. Kolesnikov, and L. Beyer, "Sigmoid loss for language image pre-training," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2023, pp. 11 975–11 986.

[6] P. K. A. Vasu, H. Pouransari, F. Faghri, R. Vemulapalli, and O. Tuzel, "MobileCLIP: Fast image-text models through multi-modal reinforced training," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024.

[7] Q. Sun, Y. Fang, L. Wu, X. Wang, and Y. Cao, "EVA-CLIP: Improved training techniques for CLIP at scale," *arXiv preprint arXiv:2303.15389*, 2023.